Physical Tail, Normalization, and Connected-Observable Certificates for the Two-Dimensional $O(3)$ Continuum Construction

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Abstract

The supplied manuscript claims a canonical interacting continuum limit of the nearest-neighbor two-dimensional $O(3)$ model, including Osterwalder–Schrader reconstruction and a gap on the entire vacuum complement. The present study separates that global construction from quantitative local deductions. Uniform exponential moments already established in the source improve its fourth-moment clipping argument to a two-point bound retaining every exponential rate strictly below the bounded-observable slab rate. Explicit positive tail sums then certify susceptibility and second-moment truncation. A physical L^2 estimate propagates their joint error through canonical normalization without counting the susceptibility twice in the field multiplier. Exact source-law and connected-moment certificates retain the conditional fluctuations lost by linear-only blocking. Original Ram moment remainders and physical metrics are used on specified finite observable spaces. None of these calculations establishes uniform renormalization, continuum existence, or a Hamiltonian gap from finite moments. Contribution assessment: C3/5, the report gives a source-specific improved two-point tail rate and explicit normalization and connected observable error certificates, conditional on the source's uniform analytic inputs and without a new global continuum or mass-gap theorem.

1 Model, normalization, and scope

On a finite periodic square lattice let $q_x \in S^2$, with normalized area measure σ, and

$$\mathrm{d}\mu_{\beta,\Lambda} = Z_{\beta,\Lambda}^{-1} \exp\left(\beta \sum_{\{x,y\}\in E(\Lambda)} q_x \cdot q_y\right) \prod_{x\in\Lambda} \mathrm{d}\sigma(q_x), \qquad \beta > 0.$$

Each unoriented bond is counted once. There is no external field, topological term, or added mass. The source [1] uses the periodic thermodynamic state μ_β, whose existence and decay at the relevant couplings are part of its analytic argument. The one-component, not vector-summed, quantities are

$$C_\beta(x) = \mathbb{E} q_0^1 q_x^1, \quad \chi_\beta = \sum_x C_\beta(x), \quad Q_\beta = \sum_x |x|^2 C_\beta(x), \quad \xi_\beta^2 = \frac{Q_\beta}{4\chi_\beta}. \tag{1}$$

The factor $4 = 2d$ is fixed by $d = 2$. Internal symmetry makes the components centered and their covariances diagonal and equal. For a vector test,

$$\Psi_\beta(f) = \frac{1}{\xi_\beta \sqrt{\chi_\beta}} \sum_{x \in \mathbb{Z}^2} f(x/\xi_\beta) \cdot q_x. \tag{2}$$

Multiplying all component covariances by three without changing the field convention would change this normalization.

The source first constructs fields $\phi(f) = Ba^2 \sum_x f(aUx) \cdot q_x$, with U an orthogonal orientation, along prescribed trajectories. Their relative two-point measure is exactly

$$S_2^{\mathrm{cut}} = \alpha_a \otimes \nu^{\mathrm{cut}}, \quad \alpha_a = a^2 \sum_x \delta_{aUx}, \quad \nu^{\mathrm{cut}} = B^2 a^2 \sum_x C_\beta(x) \delta_{aUx}. \tag{3}$$

In the continuum the first factor is Lebesgue measure. Put

$$m = \nu(\mathbb{R}^2), \qquad Q = \int |z|^2 \, \mathrm{d}\nu(z), \qquad \ell^2 = Q/(4m).$$

All normalization results below require $0 < m, Q < \infty$. The source proves positivity of Q from correlations of separated cells, not merely from a nonzero coincident covariance. Its stronger global assertions concern all real $\beta \to \infty$, the full hierarchy, and the entire vacuum complement. Those quantifiers cannot be certified by a finite matrix of moments.

2 Recovering the slab decay rate

This local improvement uses two specific source inputs: uniform exponential moments for unit-box field sums, and slab mixing for bounded local observables. It changes the passage between them, not their renormalization-theoretic proofs.

Theorem 2.1 (Exponential-moment clipping)**.** *Let X_T, Y_T be real random variables on one probability space for each $T \geq T_0 \geq 0$. Suppose constants $a, K, M, c > 0$, independent of the member of the family and of T, satisfy*

$$\mathbb{E}e^{a|X_T|}, \ \mathbb{E}e^{a|Y_T|} \leq K$$

and, for every $D \geq 0$, with $h_D(x) = \max(-D, \min(x, D))$,

$$|\operatorname{Cov}(h_D(X_T), h_D(Y_T))| \leq MD^2 e^{-cT}.$$

Then

$$|\operatorname{Cov}(X_T, Y_T)| \leq \frac{4K}{a^2} e^{-aD/2} + MD^2 e^{-cT} \quad (D \geq 0). \tag{4}$$

In particular, the choice $D = 2cT/a$ gives

$$|\operatorname{Cov}(X_T, Y_T)| \leq \frac{4}{a^2}(K + Mc^2T^2)e^{-cT}. \tag{5}$$

For every $0 < c' < c$, the right side is at most

$$\frac{4}{a^2}\left(K + \frac{4Mc^2}{e^2(c-c')^2}\right) e^{-c'T}.$$

Proof. Markov's inequality gives $\mathbb{P}(|X_T| > u) \le Ke^{-au}$. The layer-cake formula yields

$$\mathbb{E}|X_T - h_D(X_T)|^2 = \int_0^\infty 2s\,\mathbb{P}(|X_T| > D + s)\,\mathrm{d}s \le \frac{2K}{a^2}e^{-aD}.$$

It also gives $\mathbb{E}X_T^2, \mathbb{E}Y_T^2 \le 2K/a^2$. Subtract the clipped covariance in two stages:

$$\mathrm{Cov}(X_T, Y_T) - \mathrm{Cov}(h_D(X_T), h_D(Y_T)) = \mathrm{Cov}(X_T - h_D(X_T), Y_T) + \mathrm{Cov}(h_D(X_T), Y_T - h_D(Y_T)).$$

Centering is an L^2 contraction, and $|h_D(X_T)| \le |X_T|$. Cauchy–Schwarz bounds the sum by $4Ka^{-2}e^{-aD/2}$. This proves (4); substitution gives (5). Finally $\sup_{T\ge 0} T^2 e^{-(c-c')T} = 4/[e^2(c - c')^2]$. □

For the source's unit-box sums, the means vanish by internal symmetry, and the odd clips also have zero means. The covariance is therefore the two-point box mass. In `fields.tex`, the moment lemma supplies uniform factorial bounds for arbitrary translated unit boxes and explicitly proves local exponential integrability. If $\mathbb{E}|X|^n \le C^n n!$ at every order, then for $0 < a < C^{-1}$, $\mathbb{E}e^{a|X|} \le (1 - aC)^{-1}$. The source obtains odd absolute moments from adjacent even moments before using this series, so no assertion about odd signed moments is substituted for an absolute bound.

The geometric separation and slab estimate in its two-point-tail lemma give the second premise with $c = c_2$, uniformly over its admitted cutoffs, bounded offsets, and three orientations. Enlarging constants handles $T < T_0$. The original proof uses fourth moments to obtain $C/D + CD^2e^{-c_2T}$, then $D = e^{c_2T/4}$. That proof is valid. Equation (5) is a quantitative improvement using a stronger moment input already available there. It does not supply c_2, improve the transfer-operator gap, or replace the uniform slab hypothesis by a finite observation.

2.1 Explicit positive tail sums

The source transfers box correlations to the relative measure through (3), positivity, a uniform positive lower bound for $\alpha_a(Q_0)$, and a bounded covering of a Minkowski sum. Those same steps transfer (5), with changed constants and a bounded shift of the center separation.

Here is an explicit summation rule once a valid bound has been supplied. Let $Q_v = v + [-1/2, 1/2)^2$, $v \in \mathbb{Z}^2$, and suppose

$$\nu(Q_v) \le A(1 + |v|^2)e^{-c|v|} \tag{6}$$

for a nonnegative measure ν, including all boxes. Let $\Omega_R = \bigcup_{|v|_\infty \le R} Q_v$, for integer $R \ge 0$. Choose a certified q with $e^{-c} \le q < 1$, and put

$$F_j(R, q) = \sum_{n=R+1}^{\infty} n^j q^n.$$

Then

$$\nu(\Omega_R^c) \le E_0 := 8A\{F_1(R, q) + 2F_3(R, q)\}, \tag{7}$$

$$\int_{\Omega_R^c} |z|^2\,\mathrm{d}\nu(z) \le E_2 := 16A \sum_{n=R+1}^{\infty} n(n + \tfrac{1}{2})^2(1 + 2n^2)q^n. \tag{8}$$

Indeed the shell $|v|_\infty = n$ has $8n$ centers, $n \le |v| \le \sqrt{2}n$, and $|z|^2 \le 2(n+1/2)^2$ on its boxes. Every term is nonnegative; no cancellation-based tail estimate is used. For exact rational evaluation,

$$F_0(R,q) = \frac{q^{R+1}}{1-q}, \qquad F_{j+1}(R,q) = q\frac{\partial}{\partial q}F_j(R,q). \tag{9}$$

Differentiation is justified by absolute local uniform convergence for $|q| < 1$. Only degrees through five are needed in (8). A rational majorant q, A, and finite moment records thus give exact rational tail certificates. The source's unspecified constants are not silently replaced by computed numbers.

3 Canonical normalization with finite errors

For a continuum field with the relative measure ν, a positive rescaling $\phi'(f) = b\phi(f(\cdot/l))$ has

$$\nu' = b^2 l^2 (D_{1/l})_* \nu, \qquad D_{1/l}(z) = z/l.$$

The factor l^2 comes from the first insertion coordinate. Thus the unique positive choice giving susceptibility and second-moment length one is

$$l = \ell = \sqrt{\frac{Q}{4m}}, \qquad b = \frac{1}{\ell\sqrt{m}} = \frac{2}{\sqrt{Q}}. \tag{10}$$

This simplification is special to the stated two-dimensional smeared field convention. It is an algebraic identity, not a new universality or scale-setting theorem.

For independently certified intervals $0 < m_- \le m \le m_+$ and $0 < Q_- \le Q \le Q_+$,

$$\frac{Q_-}{4m_+} \le \ell^2 \le \frac{Q_+}{4m_-}, \qquad \frac{2}{\sqrt{Q_+}} \le b \le \frac{2}{\sqrt{Q_-}}. \tag{11}$$

These are the exact ranges over the rectangular set of possible (m, Q). Correlated moment information can reduce that set, but cannot justify a smaller range without an additional argument. In particular, estimating $1/\ell$ and $1/\sqrt{m}$ independently would count the mass uncertainty twice in b.

3.1 A physical test-norm bound

Assume an actual centered $O(3)$-invariant translation-invariant field Ψ has a nonnegative relative two-point measure $\bar{\nu}$ of mass one. The source's canonical limit has this property if its construction is accepted. For real vector $f \in C_c^\infty(\mathbb{R}^2;\mathbb{R}^3)$,

$$\mathbb{E}|\Psi(f)|^2 = \iint f(y)\cdot f(y+z)\,\mathrm{d}y\,\mathrm{d}\bar{\nu}(z) \le \|f\|^2_{L^2(\mathbb{R}^2;\mathbb{R}^3)}. \tag{12}$$

For each z, translation is an L^2 isometry and Cauchy–Schwarz bounds the inner integral by $\|f\|_2^2$. Integrating the positive probability measure proves the inequality. Positive definiteness alone would not give the same proof with the mass of a signed measure; positivity is an explicit hypothesis.

Suppose positive retained moments m_R, Q_R obey

$$m = m_R + \varepsilon_0, \quad Q = Q_R + \varepsilon_2, \qquad 0 \le \varepsilon_i \le E_i.$$

For truncation on Ω_R, the bounds above supply E_0, E_2. Define $\ell_R^2 = Q_R/(4m_R)$, $b_R = 2/\sqrt{Q_R}$, and $\Psi_R(f) = b_R\phi(f(\cdot/\ell_R))$. Both fields are evaluated under the *same* underlying law. Set

$$r = \frac{m_R}{m}, \quad t = \frac{Q_R}{Q}, \quad s = \frac{\ell_R}{\ell} = \sqrt{\frac{t}{r}}.$$

The exact field identity is

$$\Psi_R(f) = t^{-1/2}\Psi(f(\cdot/s)). \tag{13}$$

It compares normalization errors, not two different Gibbs measures.

Theorem 3.1 (Tail-to-normalized-field certificate)**.** *Under these assumptions, let*

$$A_0 = \sqrt{1 + E_0/m_R}, \quad D_0 = \tfrac{1}{2}\max\{\log(1 + E_0/m_R), \log(1 + E_2/Q_R)\}.$$

For $\mathcal{D}f = f + x \cdot \nabla f$, componentwise,

$$\|\Psi_R(f) - \Psi(f)\|_{L^2(\mathbb{P})} \le (A_0 - 1)\,\|f\|_2 + A_0 D_0\,\|\mathcal{D}f\|_2\,. \tag{14}$$

Proof. The dilation $U_s f = s^{-1} f(\cdot/s)$ is unitary on $L^2(\mathbb{R}^2)$. Differentiating in $u = \log s$ gives $\frac{\mathrm{d}}{\mathrm{d}u} U_{e^u} f = -U_{e^u}\mathcal{D}f$. The fundamental theorem of calculus and unitarity imply $\|U_s f - f\|_2 \le |\log s|\,\|\mathcal{D}f\|_2$. Since $t^{-1/2}s = r^{-1/2}$, (13) and (12) give

$$\|\Psi_R(f) - \Psi(f)\|_{L^2} \le (r^{-1/2} - 1)\,\|f\|_2 + r^{-1/2}|\log s|\,\|\mathcal{D}f\|_2\,.$$

Here $r^{-1/2} \le A_0$. Both $-\log r$ and $-\log t$ are nonnegative, bounded by the corresponding logarithms in the definition of D_0. The absolute difference of two numbers in these intervals is bounded by their larger endpoint, not their sum. Hence $|\log s| \le D_0$, proving the result. □

The scalar multiplier in (10) depends only on Q. After unitary transport of the test, however, the L^2 amplitude depends on m, and Q/m controls the dilation. These are compatible statements, not competing normalizations. The bound is ineffective if the retained mass or second moment has no positive lower certificate. Nor does an L^2 bound alone control a fourth cumulant; higher integrability or the composite-observable bounds below are needed.

3.2 Finite cutoffs and polynomial tests

At a fixed cutoff put $a = \xi_\beta^{-1}$ and $p_x = C_\beta(x)/\chi_\beta$. Provided the positive summable covariance is known, $\sum p_x = 1$, and the discrete counterpart is

$$\mathbb{E}|\Psi_\beta(f)|^2 \le a^2 \sum_x |f(ax)|^2. \tag{15}$$

Indeed, expand the covariance in the displacement $x - y$, and apply Cauchy–Schwarz to each translated sequence. Translation by a lattice vector preserves the sampled norm. A general dilation does not. Thus the exact identity (13) still permits a sampled-test error certificate, but the continuum unitary proof of Theorem 3.1 must not be used on a discrete grid without a quadrature or interpolation estimate.

For a finite real test bank $f = \sum_\alpha c_\alpha \varphi_\alpha$, the physical upper metric is

$$G_{\alpha\gamma} = \int \varphi_\alpha(x) \cdot \varphi_\gamma(x)\,\mathrm{d}x \quad \text{or} \quad G^a_{\alpha\gamma} = a^2 \sum_x \varphi_\alpha(ax) \cdot \varphi_\gamma(ax).$$

It bounds field variance by $c^{\mathsf{T}} G c$, not by an unweighted coefficient norm. Linearly dependent tests are quotiented out before inversion. With complex tests the first factor is conjugated and $c^* G c$ is used. The original physical-metric derivation [7] also separates a projected residual from its orthogonal omitted part; neither disappears because a finite matrix is well conditioned.

The local-moment remainder in [5] gives a concrete approximation option. On a convex cell of radius h, Taylor expansion through degree p, with every segment in the cell and $\sup \|D^{p+1} f\| \le M$, has pointwise error at most $Mh^{p+1}/(p+1)!$. Sum its squared bound times cell area, or use the exact sampled areas, and then (12) or (15). Piecewise polynomial tests need not be smooth, but the estimate extends by completion whenever the actual field is defined continuously in that norm. This is an approximation of an observable under an unchanged law. It is not a Taylor replacement of a positive Gibbs density, nor an inverse RG theorem.

4 The exact blocking law and its lost fluctuations

Let μ be an actual probability law, $K(\,\mathrm{d}V \mid q)$ a normalized positive Markov kernel, and $\pi(\,\mathrm{d}q,\,\mathrm{d}V) = \mu(\,\mathrm{d}q)K(\,\mathrm{d}V \mid q)$. For any integrable source observable $X(q)$, conditional expectation is with respect to this joint law. Wherever the source generating functions exist,

$$\mathbb{E}_\mu e^{z\cdot X} = \mathbb{E}_{\mu K}\left[\mathbb{E}_\pi(e^{z\cdot X} \mid V)\right]. \tag{16}$$

The inner quantity is generally not $\exp(z \cdot \mathbb{E}[X \mid V])$. At the origin its logarithm has conditional covariance as its quadratic coefficient. In particular

$$\operatorname{Cov}(X) = \mathbb{E}\operatorname{Cov}(X \mid V) + \operatorname{Cov}(\mathbb{E}[X \mid V]). \tag{17}$$

These identities follow from the tower property, or by expanding $X = \mathbb{E}[X \mid V] + (X - \mathbb{E}[X \mid V])$ and using conditional orthogonality. They do not require a cumulant series away from the origin.

The source's kernel is normalized exactly:

$$K_b(\,\mathrm{d}V \mid q) = \frac{e^{-b|V-\widehat{m}(q)|^2/2}}{z_2(b)}\,\mathrm{d}\sigma(V), \qquad z_2(b) = \frac{1-e^{-2b}}{2b},$$

with its specified tagged definition of $\widehat{m}$ even when the block average is small. The source keeps all higher source degrees, including spin-independent terms, and extracts only a source-independent bulk scalar. Individual terms of its signed cover expansion are not probability laws to which (17) can be applied separately.

Proposition 4.1 (An exact one-spin diagnostic)**.** *Let q be uniform on S^2, and observe V with the preceding kernel with $\widehat{m} = q$, $b > 0$. Then V is uniform and*

$$\mathbb{E}[q \mid V] = L(b)V, \qquad L(b) = \coth b - b^{-1}.$$

For the first coordinate, replacing the full conditional source by its linear conditional mean changes the variance from $1/3$ to $L(b)^2/3$ and the fourth cumulant from $-2/15$ to $-2L(b)^4/15$.

Proof. The joint density is proportional to $e^{bq\cdot V}$, symmetric in q, V, and invariant under simultaneous rotations. Its marginal normalizer is $Z(b) = \sinh(b)/b$. Differentiating gives $\mathbb{E}(q \cdot V \mid V) = Z'(b)/Z(b) = L(b)$; axial symmetry determines the full conditional mean. A coordinate of a uniform sphere point has uniform law on $[-1,1]$, so its second and fourth moments are $1/3, 1/5$. The cumulant is $1/5 - 3(1/3)^2 = -2/15$. Scaling the coordinate by $L(b)$ proves the assertions. □

For finite b, $0 < L(b) < 1$, since the conditional distribution has positive density and nonzero spread. Thus the omitted variance $(1 - L(b)^2)/3$ is genuinely positive. This single-site calculation diagnoses source bookkeeping; it is not the multi-site RG map. A purely rational diagnostic law is

$$\mathrm{d}\pi(q,V) = (1 + 3\lambda q \cdot V)\,\mathrm{d}\sigma(q)\,\mathrm{d}\sigma(V), \qquad |\lambda| \le 1/3.$$

It is positive and normalized, and $\mathbb{E}(q \mid V) = \lambda V$, by $\int q_i q_j \, d\sigma = \delta_{ij}/3$. All polynomial moments of this diagnostic can be checked exactly. It must not be substituted for the source's exponential kernel.

5 Connected-observable certificates

For centered real $X_1, \dots, X_4 \in L^4(\mathbb{P})$, set

$$C_{ij} = \mathbb{E}X_i X_j, \qquad \kappa_4 = \mathbb{E}X_1X_2X_3X_4 - C_{12}C_{34} - C_{13}C_{24} - C_{14}C_{23}.$$

Centering is a hypothesis, satisfied by the source's component fields by internal symmetry. Finite estimates below are not assertions that four specified continuum cells have already been constructed.

Theorem 5.1 (Source-disc and alignment budget)**.** *Suppose $X_1, \dots, X_4$ are centered real variables and $Y_1, \dots, Y_4$ are first coordinates of sphere-valued variables under an internally rotation-invariant law. Suppose their generating functions are holomorphic on a neighborhood of the closed complex polydisc $|z_i| \le \rho$, and differ there by at most η. Assume certified numbers $\epsilon_i \ge 0$, with $\epsilon_1 = 0$, satisfy $\|Y_i - Y_1\|_2 \le \epsilon_i$. Put*

$$a = \eta\rho^{-2}, \quad b = \eta\rho^{-4}, \quad S = \sum_i \epsilon_i, \quad d_{ij} = a + \epsilon_i + \epsilon_j.$$

Then

$$\left|\kappa_4(X) + \frac{2}{15}\right| \le b + 2S + 2a + d_{12}d_{34} + d_{13}d_{24} + d_{14}d_{23}. \tag{18}$$

In particular a right side strictly smaller than $2/15$ certifies a negative fourth cumulant.

Proof. Cauchy's coefficient formula gives errors at most a, b for mixed second and fourth moments with distinct source indices. There is no additional factorial since each index occurs once. Since $|Y_i| \le 1$, telescoping products and Cauchy–Schwarz give

$$|\mathbb{E}Y_iY_j - 1/3| \le \epsilon_i + \epsilon_j, \qquad |\mathbb{E}Y_1Y_2Y_3Y_4 - 1/5| \le S.$$

The marginal of the first sphere variable is uniform, supplying $1/3, 1/5$. Consequently $|C_{ij} - 1/3| \le d_{ij}$ and the fourth moment differs from $1/5$ by at most $b + S$. For every disjoint pairing $(ij), (kl)$,

$$|C_{ij}C_{kl} - 1/9| \le (d_{ij} + d_{kl})/3 + d_{ij}d_{kl}.$$

The three linear pairing contributions sum to $2a + S$. Adding the fourth-moment error proves (18). □

Here is a direct input from the source, not a fitted value. Its localized comparison on D terminal sites, at component source radius one, supplies

$$\eta_D \le e^{3D}(e^{DR_0} - 1) + e^{3D+DR_0}(e^{Dw_0} - 1). \tag{19}$$

For four distinct cells use $D = 4, \rho = 1$; four scalar sources embed in the source's larger vector-source polydisc. A path of h_i terminal edges between cell i and cell 1, together with a verified edge estimate $\mathbb{E}|V_x - V_y|^2 \le v$, gives $\epsilon_i \le h_i\sqrt{v}$ by the L^2 triangle inequality. In the source, v is bounded using the gradient threshold and its bad-bond probability. All R_0, w_0, v and path lengths must be valid uniformly in the claimed cutoff and volume family.

For example take certified $a = b = 1/1000$ and $\epsilon_2 = \epsilon_3 = \epsilon_4 = 1/1000$. The right side of (18) is $9000/10^6 + 18/10^6 = 4509/500000 < 2/15$. This is an exact budget fixture, not a claim that the source supplies these numerical constants at a specified H. Passage to separated smooth continuum tests additionally uses the source's boundary-strip estimate $C(\varepsilon + a_N)$, its moment convergence, and an error budget for the smoothing. The cell cumulant cannot bypass those steps.

5.1 A common-bank certificate with quadratic remainder

The original common-law covariance calculation in [6] emphasizes using one actual law for all observables and keeping the remainder correlated with the approximation. The following version is tailored to a connected four-point observable.

Proposition 5.2 (Projected pair products)**.** *Let $U_{ij} = X_iX_j - C_{ij}$. Let $\psi = (\psi_1, \dots, \psi_s)^{\mathsf{T}}$ be a centered real L^2 bank under the same law, with positive definite physical Gram matrix $G = \mathbb{E}\psi\psi^{\mathsf{T}}$. Set*

$$h_{ij} = \mathbb{E}(\psi U_{ij}), \qquad e_{ij}^2 = \mathbb{E}U_{ij}^2 - h_{ij}^{\mathsf{T}}G^{-1}h_{ij}.$$

Then $e_{ij}^2 \geq 0$, and

$$\left|\kappa_4 - \left(h_{12}^{\mathsf{T}}G^{-1}h_{34} - C_{13}C_{24} - C_{14}C_{23}\right)\right| \leq e_{12}e_{34}. \tag{20}$$

The two other pairings give two further valid intervals for the same number, whose intersection is also valid.

Proof. The orthogonal projection of U_{ij} onto the bank is $\psi^{\mathsf{T}}G^{-1}h_{ij}$. Its residual r_{ij} is orthogonal to every bank element and has norm e_{ij}. Therefore $\mathbb{E}U_{12}U_{34} = h_{12}^{\mathsf{T}}G^{-1}h_{34} + \mathbb{E}r_{12}r_{34}$. But $\mathbb{E}U_{12}U_{34} = \kappa_4 + C_{13}C_{24} + C_{14}C_{23}$. Cauchy–Schwarz gives (20). □

The quadratic product of residuals is the benefit of a common exact orthogonal projection. Approximate coefficients, approximate Gram inversion, or different probability laws require additional linear error terms. Repeated fourth moments $\mathbb{E}X_i^2X_j^2$, cross moments with the bank, and all pair covariances are necessary inputs. A positive semidefinite fitted matrix alone does not certify that these moments belong to the source's Gibbs law. For singular G, first pass to an independent bank in its physical quotient. No bound for an omitted infinite-dimensional subspace follows from invertibility on the quotient.

For an exact example let all $X_i = q^1$ for one uniform sphere point, and take the single feature $\psi = (q^1)^2 - 1/3$. Then $G = h_{ij} = 4/45$ and $e_{ij} = 0$, so the certificate returns $4/45 - 2/9 = -2/15$. Replacing ψ by $\tau\psi$, $\tau \neq 0$, changes G by τ^2 and h by τ, but leaves the physical answer unchanged. This demonstrates metric invariance and a zero projection remainder, not spatial separation or a Hamiltonian gap.

6 The source architecture and its analytic obligations

The source is a substantial construction rather than a finite-moment argument. The following map records what the local results leave untouched. File names refer to its pinned build, and the complete literal interval inventory accompanies the report.

Initialization and preliminary mixing. `introduction`, `history`, and `method-history` state the claims and place them in the RG literature. `setup` fixes the probability kernel, coupling bands, graph domains, canonical slots, regular-error norms, complete covering records, and order of parameter choices. `preliminary` uses pinned-square Ward identities, an augmented current variance, a stiffness iteration, and conditional ferromagnetic decompositions. Its decisive estimate is a preliminary length at most $C\exp((4\pi - \eta)\beta)$ for some $\eta > 0$, uniform under weakening or deletion of bonds. The strict gain η, not merely eventual decay at each coupling, is used in the later reference-box construction. The stiffness matrix is not assumed positive; adding auxiliary Gaussian currents and computing their total variance is distinct from declaring it positive by a finite fit.

Free kernels and the exact nonlinear step. `free` constructs massless scalar precisions, conditional Gaussian means, edge lifts, and positive kinetic completions for ordinary blocks and the two 3-4-5 inclined blocks. Adjoints retain the fine-cell factor l^2; inclined aliases use their actual period lattice. Common-history differences contract like $(A_0/L^2)^r$ on a fixed complex strip. `rg` then treats the nonlinear sphere law, not just these free matrices. Complete exceptional records contain all tested sites, predicates, and connecting paths, including negative eligibility tests. The exact energy accounting, protected chart domains, integration over joint Gaussian factors, and convergent covering sums remain necessary. The normalized auxiliary scalar removes one of three ambient colors, leaving two physical tangent colors. It is not an added physical field.

Regular errors have eight derivatives on their specified graph domains, whereas their differences use seven. A pointwise remainder does not certify these derivatives. The source does not differentiate a hard mask jump or extrapolate a local chart through the antipodal singularity. It retains all remainder and winding terms after finite canonical Taylor extraction. Finite polynomial closure would not establish that exact step.

Couplings, densities, and source insertions. `shooting` obtains $\beta_N = H + (\log L)N/(2\pi) + O_{L,H}(\log(2+N))$ and matches endpoints using a two-boundary estimate. The first kinetic increment is calibrated through finite-volume Laplace coefficients, with uniformity of coefficients distinguished from a uniform bare asymptotic remainder. `comparison` integrates differences of complete covering sums before estimating them. The base law is positive and has the required reflection positivity; individual signed activities are not divided into probabilities. Its replacement operation retains the original region geometry, and every forest retains its complete inventories. These requirements justify the volume-dependent error $C_H v L^{-(2-\upsilon)K}$.

`sources` substitutes sources exactly and keeps every positive source degree. The factor $B_N = \prod_{j=1}^{N} c_j^{-1}$ is separate from the area factor. Its allowed growth, $\log B_N = O(1 + N^{0.9})$, is not a uniform bound on B_N. Statements about a finite terminal source set have constants depending on that set; they do not automatically hold uniformly over a growing set.

Volume, continuum, and all bare couplings. `trace` combines a fixed physical reference-box test with positivity of transfer operators in both directions. Excited-trace control at one circumference alone would not control all larger circumferences. Shifted temporal closing conditions give the inclined-torus comparison. `continuum` first fixes a torus and coarse cell mesh, then removes the cutoff and refines the tests. Bounded characteristic functions, not unbounded field moments, compare increasing tilted volumes. Uniform factorial moments subsequently give uniform integrability. Boundary-strip control permits moving cell indicators.

`trajectories` supplies bounded-offset admission and summable kinetic sensitivity, including bulk data even in a finite-period comparison. `volume` makes the reference-box and slab estimates uniform over these offsets. `fields` derives factorial moments, relative-measure tails, positive susceptibility and second moment, and canonical normalization. `uniqueness` uses continuous offset families and two scale relations: the ordinary factor L, a power of two, and the inclined factor 5. Positive lengths force incommensurability of the corresponding offset shifts. Together with the irrational angle having cosine 3/5, these relations remove both orientation and offset dependence. A finite list of offsets or rotation angles does not prove this step.

Reconstruction and the full gap. `os` verifies the specified Euclidean axioms and growth conditions for the reconstruction theorem used there. Its polynomial clustering argument already uses exponential integrability, unlike the earlier fourth-moment two-point clipping argument refined

here. The source proves density of strictly time-ordered product states in the reflection norm, carefully not claiming Schwartz density after cutting equal-time surfaces. For each centered vector in this dense class, a common exponential rate excludes low-energy mass in its positive spectral measure; bounded spectral projections then extend the exclusion to the closure. This is a full-space argument. A nonzero separated fourth cumulant proves a failure of Wick factorization, but cannot supply the dense-class spectral argument on its own.

For example, if a finite observable bank sees only the first s coordinates of ℓ^2, the operators with eigenvalues $1, \dots, 1, 1, 1, \dots$ and $1, \dots, 1, 1/2, 1/3, \dots$ have identical compressions to that bank. The second has no positive lower bound on the omitted subspace. This elementary obstruction explains why the physical Gram calculations in this report do not certify the source's gap.

6.1 Imported inputs, retained details, and corrections

The source repeatedly refers to its $O(4)$ companion for scalar localization, Gaussian product and tree estimates, Ward-identity bookkeeping, and finite-volume coefficient calculations. The present report reads the $O(3)$ manuscript's own adaptations but does not freshly audit that entire companion or every cited classical proof. Those deep inputs, their uniform constants, and the complete nonlinear closure remain imported for the local application. Reading a reference-locator table is not reading the proof to which it points.

No substantive error in the inspected source chain is established here. In particular, the fourth-moment tail estimate is valid; it is improved, not repaired. The factor 4 in second-moment length, the area factors l^2 and 25, the one-component susceptibility, the retained higher conditional source terms, the positive complete density, and the order of cutoff and volume limits have all been preserved. The exact cancellation $b = 2/\sqrt{Q}$ simplifies error propagation but does not alter the original canonical normalization.

7 Original Ram connections and public context

The original local-moment chapter [5] proves Taylor-integral remainder estimates with absolute moments and derivatives on the whole relevant segment. Its noncompact-kernel section explicitly separates truncation from renormalization of the truncated kernel. That separation motivates Sections 2.1–3: tail mass, second-moment loss, and rescaling are distinct errors. Its warnings about positivity and finite-model inversion apply directly to any attempted finite polynomial replacement of a Gibbs kernel.

The common-law covariance derivations in [6] retain between-component covariance for mixtures and conditional laws, and use correlated L^2 remainders instead of independent noise. This is the relevant original connection for (17) and Proposition 5.2. The physical-metric chapter [7] makes coefficient Gram norms and omitted residuals explicit. Here these ideas are applied to smeared spin fields and centered pair products, not credited as a new theory of projections or probability moments.

Public primary records were refreshed on October 9, 2026; the cached manuscript was authenticated against the captured October 8 tree. Aru, Garban, and Sepúlveda [2] give rigorous high-inverse-temperature transverse GFF behavior and related percolation results. This local-fluctuation statement is not the canonical all-scale interacting limit asserted in the supplied manuscript. Dybalski, Stottmeister, and Tanimoto [3] analyze the constrained variational step of a two-dimensional $O(4)$ model, using a contraction argument and lattice Green-function control; it is not a full $O(3)$ continuum construction. Aru and Korzhenkova [4], in the September 2025 revision, take spin dimension to infinity before torus size. That order is different from fixing three components and

removing a spatial cutoff. These records were consulted for their primary abstracts and version metadata, not independently proof-audited.

The supplied manuscript's strong global claim is therefore treated as its claim, neither accepted merely because a file is publicly available nor disproved by the narrower scope of earlier work. No private corpus material was uploaded or used as a public search query. None of the local arguments needs a global continuum theorem imported from another report in this collection.

8 Usability, costs, and further work

Certificate	Advantage	Required cost or limitation
Exponential clipping	Retains every rate below the bounded slab rate	Uniform exponential moments and actual bounded-observable mixing
Positive shell sums	Exact rational mass and second-moment budgets	Valid all-box majorants, not sampled decay
Canonical physical norm	Separates amplitude and dilation without duplicate mass error	Positive retained m_R, Q_R; actual common law
Source-disc budget	Explicit sign margin for a connected fourth moment	Uniform analytic comparison and endpoint alignment
Pair-product projection	Quadratic residual product; invariant under basis rescaling	Actual cross and repeated fourth moments; certified Gram inverse

The tail certificate becomes expensive when a usable exponent is small or its prefactor is large. Requiring a pure rate very close to c also enlarges the displayed constant like $(c - c')^{-2}$. Near-zero Q_R destabilizes scale setting. A larger observable bank can reduce projection residuals but increases moment acquisition, inversion, and interval-conditioning costs. These are explicit tradeoffs, not asserted computational speedups.

A feasible next step is to supply independently validated numerical enclosures for a, K, c, A in a specified admitted family, then choose the smallest retained radius meeting a requested physical test tolerance. Another is to combine exact composite projections with source-disc bounds on the missing repeated moments. Any continuum use still needs uniformity in the cutoff. Extending these estimates to a signed relative measure would require its total variation, and loses the positive monotone moment intervals. Extending to a different dimension changes both the rescaling Jacobian and the dilation generator; the two-dimensional cancellation cannot be copied unchanged.

9 Evidence and contribution assessment

The entire set of 23 cached text files was literally displayed in bounded chunks: all scientific modules, the main file, bibliography, reference locators, README, and two figure sources. Figure code and captions were read, not independently rendered as source illustrations. One combined display was truncated in the middle of `free.tex`; that whole affected interval was redisplayed untruncated before being accepted as read. This is complete textual reading of this cached manuscript, not formal verification of all its analytic estimates or external references.

Fresh private original reading covers the local-moment chapter lines 1–446 and the physical-metric chapter lines 1–210, plus TR-core-pat-16 lines 1–40 and 216–328. The latter's remaining lines and the chapters included by input commands in the moment chapter were not read. No whole-book or whole-corpus scope is claimed. The accompanying ledger records literal intervals, unread complements, and matching before/after hashes of the existing local files, without claiming unobserved historical identity.

The independent exact-check program tests rational tail identities, moment-normalization cancellation and interval endpoints, positive one-spin diagnostic moments, the cumulant error algebra, common-bank residual identities, and finite discrete physical-norm inequalities. Its JSON records parameter cases separately from assertions. These finite tests supplement the proofs above; they are not numerical integration of the original Gibbs model, a run of its RG transformation, or a global continuum certificate. Independent review fixtures are not included in the author test count.

The contribution is assessed at C3, independently of applicability L3. The notable local increment is the improved source-specific two-point tail passage, coupled to explicit positive-tail and canonical physical-norm error propagation, and connected-observable budgets preserving the source law. Classical clipping, Cauchy's formula, conditional expectation, and orthogonal projection are credited as tools, not inventions. Neither a new global $O(3)$ existence theorem nor a new mass-gap proof is claimed.

References

[1] OpenAI. *The canonical massive continuum limit of the two-dimensional $O(3)$ model*. October 4, 2026, pinned supplied manuscript. Scientific modules in its build directory; captured-tree provenance and local hashes accompany this report.

[2] J. Aru, C. Garban, and A. Sepúlveda. *Percolation for 2D classical Heisenberg model and exit sets of vector valued GFF*. Primary preprint record, 2022. https://arxiv.org/abs/2212.06767.

[3] W. Dybalski, A. Stottmeister, and Y. Tanimoto. *The Bałaban variational problem in the non-linear sigma model*. Primary preprint record, 2024. https://arxiv.org/abs/2403.09800.

[4] J. Aru and A. Korzhenkova. *The Spherical model and the large-N limit of the Spin $O(N)$ model via the Gaussian free field*. Version 2, September 3, 2025. https://arxiv.org/abs/2405.04501.

[5] M. Subbarao. RamCalBooks, Volume 1, revision 2, Chapter 3, local moments. Confidential original chapter; `ch03-local-moments-rev2.tex`, lines 1–446.

[6] M. Subbarao. *Joint-Moment Bayesian Neural Inference and Inverse Uncertainty Propagation*, TR-core-pat-16. Confidential original; selected proof passages, lines 216–328.

[7] M. Subbarao. RamCalBooks, Volume 21, physical metrics chapter. Confidential original; `ch01d-physical-metrics.tex`, lines 1–210.